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Indefinite Integrals

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	91 questions
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MCQ (Single Correct Answer) 91 questions

Q1. JEE Main 2026 (Online) 2nd April Evening Shift

Let $f(x)=\int\left(\frac{16x+24}{x^2+2x-15}\right)dx$. If $f(4)=14\log_e(3)$ and $f(7)=\log_e(2^\alpha\cdot 3^\beta)$, $\alpha,\beta\in\mathbb{N}$, then $\alpha+\beta$ is equal to :

(A) 31 (B) 37 (C) 39 (D) 41

Open solution

Q2. JEE Main 2026 (Online) 28th January Evening Shift

Let $f(x)=\int\frac{dx}{x^{2/3}+2x^{1/2}}$, be such that $f(0)=-26+24\log_e(2)$. If $f(1)=a+b\log_e(3)$, where $a,b\in\mathbb{Z}$, then $a+b$ is equal to :

(A) -5 (B) -11 (C) -18 (D) -26

Open solution

Q3. JEE Main 2026 (Online) 28th January Morning Shift

If $\int\left(\frac{1-5\cos^2 x}{\sin^5 x\cos^2 x}\right)dx=f(x)+\mathrm{C}$, where C is the constant of integration, then $f\left(\frac{\pi}{6}\right)-f\left(\frac{\pi}{4}\right)$ is equal to

(A) $\frac{1}{\sqrt{3}}(26-\sqrt{3})$ (B) $\frac{4}{\sqrt{3}}(8-\sqrt{6})$ (C) $\frac{1}{\sqrt{3}}(26+\sqrt{3})$ (D) $\frac{2}{\sqrt{3}}(4+\sqrt{6})$

Open solution

Q4. JEE Main 2026 (Online) 24th January Morning Shift

Let $f(t)=\int\left(\frac{1-\sin(\log_e t)}{1-\cos(\log_e t)}\right)dt, t>1$.

If $f\left(e^{\pi/2}\right)=-e^{\pi/2}$ and $f\left(e^{\pi/4}\right)=\alpha e^{\pi/4}$, then α equals

(A) $1+\sqrt{2}$ (B) $-1-2\sqrt{2}$ (C) $-1-\sqrt{2}$ (D) $-1+\sqrt{2}$

Open solution

Q5. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\mathrm{I}(x)=\int\frac{3dx}{(4x+6)\left(\sqrt{4x^2+8x+3}\right)}$ and $\mathrm{I}(0)=\frac{\sqrt{3}}{4}+20$. If

$\mathrm{I}\left(\frac{1}{2}\right)=\frac{a\sqrt{2}}{b}+\mathrm{c}$, where $a,b,\mathrm{c}\in\mathrm{N}, \gcd(a,b)=1$, then $a+b+c$ is equal to :

(A) 30 (B) 29 (C) 28 (D) 31

Open solution

Q6. JEE Main 2026 (Online) 23rd January Morning Shift

Let $f(x)=\int\frac{\left(2-x^2\right)\cdot \mathrm{e}^x}{(\sqrt{1+x})(1-x)^{3/2}}\,\mathrm{d}x$. If $f(0)=0$, then $f\left(\frac{1}{2}\right)$ is equal to:

(A) $\sqrt{2\mathrm{e}}-1$ (B) $\sqrt{2\mathrm{e}}+1$ (C) $\sqrt{3\mathrm{e}}-1$ (D) $\sqrt{3\mathrm{e}}+1$

Open solution

Q7. JEE Main 2025 (Online) 3rd April Morning Shift

Let $f(x)=\int x^3\sqrt{3-x^2}dx$. If $5f(\sqrt{2})=-4$, then $f(1)$ is equal to

(A) $-\frac{6\sqrt{2}}{5}$ (B) $-\frac{8\sqrt{2}}{5}$ (C) $-\frac{2\sqrt{2}}{5}$ (D) $-\frac{4\sqrt{2}}{5}$

Open solution

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Q8. JEE Main 2025 (Online) 28th January Evening Shift

If $f(x)=\int \frac{1}{x^{1/4}(1+x^{1/4})}dx$, $f(0)=-6$, then $f(1)$ is equal to :

(A) $4\left(\log_e 2-2\right)$ (B) $\log_{e^2} 2+2$

(C) $2-\log e^2$ (D) $4\left(\log_e 2+2\right)$

Open solution

Q9. JEE Main 2025 (Online) 23rd January Evening Shift

Let $\int x^3 \sin x\, dx = g(x)+C$, where C is the constant of integration. If $8\left(g\left(\frac{\pi}{2}\right)+g'\left(\frac{\pi}{2}\right)\right)=\alpha\pi^3+\beta\pi^2+\gamma, \alpha,\beta,\gamma\in Z$, then $\alpha+\beta-\gamma$ equals :

(A) 47 (B) 55 (C) 62 (D) 48

Open solution

Q10. JEE Main 2025 (Online) 23rd January Morning Shift

Let $I(x)=\int \frac{dx}{(x-11)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}}$. If $I(37)-I(24)=\frac{1}{4}\left(\frac{1}{b^{\frac{1}{13}}}-\frac{1}{c^{\frac{1}{13}}}\right), b,c\in\mathcal{N}$, then $3(b+c)$ is equal to

(A) 39 (B) 22 (C) 40 (D) 26

Open solution

Q11. JEE Main 2025 (Online) 22nd January Evening Shift

If $\int e^x\left(\frac{x\sin^{-1}x}{\sqrt{1-x^2}}+\frac{\sin^{-1}x}{(1-x^2)^{3/2}}+\frac{x}{1-x^2}\right)dx=g(x)+C$, where C is the constant of integration, then $g\left(\frac{1}{2}\right)$ equals :

(A) $\frac{\pi}{6}\sqrt{\frac{e}{3}}$ (B) $\frac{\pi}{6}\sqrt{\frac{e}{2}}$

(C) $\frac{\pi}{4}\sqrt{\frac{e}{3}}$ (D) $\frac{\pi}{4}\sqrt{\frac{e}{2}}$

Open solution

Q12. JEE Main 2024 (Online) 9th April Morning Shift

Let $\int \frac{2-\tan x}{3+\tan x}\,dx=\frac{1}{2}(\alpha x+\log_e|\beta\sin x+\gamma\cos x|)+C$, where C is the constant of integration. Then $\alpha+\frac{\gamma}{\beta}$ is equal to :

(A) 3 (B) 7 (C) 1 (D) 4

Open solution

Q13. JEE Main 2024 (Online) 8th April Morning Shift

Let $I(x)=\int \frac{6}{\sin^2 x(1-\cot x)^2}dx$. If $I(0)=3$, then $I\left(\frac{\pi}{12}\right)$ is equal to

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) $6\sqrt{3}$ (D) $3\sqrt{3}$

Open solution

Q14. JEE Main 2024 (Online) 6th April Evening Shift

If $\int \frac{1}{a^2\sin^2 x+b^2\cos^2 x}\,dx=\frac{1}{12}\tan^{-1}(3\tan x)+$ constant, then the maximum value of $a\sin x+b\cos x$, is :

(A) $\sqrt{41}$ (B) $\sqrt{39}$ (C) $\sqrt{40}$ (D) $\sqrt{42}$

Open solution

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Q15. JEE Main 2024 (Online) 29th January Evening Shift

If $\int \frac{\sin^{\frac{3}{2}} x+\cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}}dx = A\sqrt{\cos\theta \tan x - \sin\theta} + B\sqrt{\cos\theta - \sin\theta \cot x} + C$, where C is the integration constant, then AB is equal to

(A) $2\sec\theta$ (B) $8\,\text{cosec}(2\theta)$ (C) $4\,\text{cosec}(2\theta)$ (D) $4\sec\theta$

Open solution

Q16. JEE Main 2024 (Online) 29th January Morning Shift

For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, if $y(x) = \int \frac{\text{cosec}\, x+\sin x}{\text{cosec}\, x \sec x+\tan x \sin^2 x}dx$, and $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^-} y(x) = 0$ then $y\left(\frac{\pi}{4}\right)$ is equal to

(A) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(C) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$

Open solution

Q17. JEE Main 2024 (Online) 27th January Evening Shift

The integral $\int \frac{(x^8-x^2)\mathrm{d}x}{(x^{12}+3x^6+1)\tan^{-1}\left(x^3+\frac{1}{x^3}\right)}$ is equal to :

(A) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)^{1/3}+\mathrm{C}$

(B) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)+\mathrm{C}$

(C) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)^{1/2}+\mathrm{C}$

(D) $\log_e\left(\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|\right)^{3}+\mathrm{C}$

Open solution

Q18. JEE Main 2023 (Online) 10th April Evening Shift

For $\alpha,\beta,\gamma,\delta \in \mathbb{N}$, if $\int \left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x dx = \frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x} - \frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x} + C$, where $e = \sum\limits_{n=0}^{\infty}\frac{1}{n!}$ and C is constant of integration, then $\alpha+2\beta+3\gamma-4\delta$ is equal to :

(A) -8 (B) -4 (C) 1 (D) 4

Open solution

Q19. JEE Main 2023 (Online) 10th April Morning Shift

If $I(x) = \int e^{\sin^2 x}(\cos x \sin 2x - \sin x)dx$ and $I(0) = 1$, then $I\left(\frac{\pi}{3}\right)$ is equal to :

(A) $-e^{\frac{3}{4}}$ (B) $-\frac{1}{2}e^{\frac{3}{4}}$ (C) $e^{\frac{3}{4}}$ (D) $\frac{1}{2}e^{\frac{3}{4}}$

Open solution

Q20. JEE Main 2023 (Online) 8th April Evening Shift

The integral $\int \left[\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x\right]\ln\left(\frac{ex}{2}\right)dx$ is equal to :

(A) $\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x+C$ (B) $\left(\frac{x}{2}\right)^x-\left(\frac{2}{x}\right)^x+C$

(C) $\left(\frac{x}{2}\right)^x\log_2\left(\frac{2}{x}\right)+C$ (D) None

Open solution

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Q21. JEE Main 2023 (Online) 8th April Morning Shift

Let $I(x)=\int \frac{(x+1)}{x\left(1+xe^{x}\right)^{2}}dx, x>0$. If $\lim\limits_{x\to\infty} I(x)=0$, then $I(1)$ is equal to :

(A) $\frac{e+1}{e+2}-\log_e(e+1)$ (B) $\frac{e+1}{e+2}+\log_e(e+1)$ (C) $\frac{e+2}{e+1}-\log_e(e+1)$ (D) $\frac{e+2}{e+1}+\log_e(e+1)$

Open solution

Q22. JEE Main 2023 (Online) 6th April Morning Shift

Let $I(x)=\int \frac{x^2(x\sec^2 x+\tan x)}{(x\tan x+1)^2}dx$. If $I(0)=0$, then $I\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\log_e\frac{(\pi+4)^2}{32}-\frac{\pi^2}{4(\pi+4)}$ (B) $\log_e\frac{(\pi+4)^2}{16}-\frac{\pi^2}{4(\pi+4)}$

(C) $\log_e\frac{(\pi+4)^2}{16}+\frac{\pi^2}{4(\pi+4)}$ (D) $\log_e\frac{(\pi+4)^2}{32}+\frac{\pi^2}{4(\pi+4)}$

Open solution

Q23. JEE Main 2023 (Online) 25th January Morning Shift

Let $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$. If $f(3)=\frac{1}{2}(\log_e 5-\log_e 6)$, then $f(4)$ is equal to

(A) $\log_e 19-\log_e 20$ (B) $\log_e 17-\log_e 18$

(C) $\frac{1}{2}(\log_e 19-\log_e 17)$ (D) $\frac{1}{2}(\log_e 17-\log_e 19)$

Open solution

Q24. JEE Main 2022 (Online) 29th July Evening Shift

For $I(x)=\int \frac{\sec^2 x-2022}{\sin^{2022}x}dx$, if $I\left(\frac{\pi}{4}\right)=2^{1011}$, then

(A) $3^{1010}I\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (B) $3^{1010}I\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

(C) $3^{1011}I\left(\frac{\pi}{3}\right)-I\left(\frac{\pi}{6}\right)=0$ (D) $3^{1011}I\left(\frac{\pi}{6}\right)-I\left(\frac{\pi}{3}\right)=0$

Open solution

Q25. JEE Main 2022 (Online) 26th July Evening Shift

The integral $\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x-\sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)}dx$ is equal to

(A) $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right|+C$

(B) $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)}\right|+C$

(C) $\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}\right|+C$

(D) $\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}-\frac{\pi}{6}\right)}\right|+C$

Open solution

Q26. JEE Main 2022 (Online) 27th June Morning Shift

If $\int \frac{(x^2+1)e^x}{(x+1)^2}dx=f(x)e^x+C$, where C is a constant, then $\frac{d^3f}{dx^3}$ at x = 1 is equal to :

(A) $-\frac{3}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

Open solution

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Q27. JEE Main 2022 (Online) 26th June Evening Shift

If $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}dx = g(x) + c, g(1) = 0$, then $g\left(\frac{1}{2}\right)$ is equal to :

(A) $\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right) + \frac{\pi}{3}$

(B) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) + \frac{\pi}{3}$

(C) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) - \frac{\pi}{3}$

(D) $\frac{1}{2}\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right) - \frac{\pi}{6}$

Open solution

Q28. JEE Main 2022 (Online) 25th June Morning Shift

Let $g : (0, \infty) \to R$ be a differentiable function such that

$\int \left(\frac{x(\cos x - \sin x)}{e^x+1} + \frac{g(x)(e^x+1-xe^x)}{(e^x+1)^2}\right)dx = \frac{x\,g(x)}{e^x+1} + c$, for all x > 0, where c is an arbitrary constant. Then :

(A) g is decreasing in $\left(0, \frac{\pi}{4}\right)$

(B) g' is increasing in $\left(0, \frac{\pi}{4}\right)$

(C) g + g' is increasing in $\left(0, \frac{\pi}{2}\right)$

(D) g − g' is increasing in $\left(0, \frac{\pi}{2}\right)$

Open solution

Q29. JEE Main 2021 (Online) 31st August Morning Shift

The integral $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}}dx$ is equal to : (where C is a constant of integration)

(A) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{1}{4}} + C$

(B) $\frac{3}{4}\left(\frac{x+2}{x-1}\right)^{\frac{5}{4}} + C$

(C) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}} + C$

(D) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{\frac{5}{4}} + C$

Open solution

Q30. JEE Main 2021 (Online) 18th March Morning Shift

The integral $\int \frac{(2x-1)\cos\sqrt{(2x-1)^2+5}}{\sqrt{4x^2-4x+6}}dx$ is equal to (where c is a constant of integration)

(A) $\frac{1}{2}\sin\sqrt{(2x-1)^2+5} + c$

(B) $\frac{1}{2}\cos\sqrt{(2x+1)^2+5} + c$

(C) $\frac{1}{2}\cos\sqrt{(2x-1)^2+5} + c$

(D) $\frac{1}{2}\sin\sqrt{(2x+1)^2+5} + c$

Open solution

Q31. JEE Main 2021 (Online) 25th February Evening Shift

The integral $\int \frac{e^{3\log_e 2x} + 5e^{2\log_e 2x}}{e^{4\log_e x} + 5e^{3\log_e x} - 7e^{2\log_e x}}dx$, x > 0, is equal to : (where c is a constant of integration)

(A) $\log_e\sqrt{x^2+5x-7} + c$

(B) $4\log_e|x^2+5x-7| + c$

(C) $\frac{1}{4}\log_e|x^2+5x-7| + c$

(D) $\log_e|x^2+5x-7| + c$

Open solution

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Q32. JEE Main 2021 (Online) 25th February Morning Shift

The value of the integral $\int \frac{\sin\theta.\sin 2\theta(\sin^6\theta+\sin^4\theta+\sin^2\theta)\sqrt{2\sin^4\theta+3\sin^2\theta+6}}{1-\cos 2\theta}\,d\theta$ is :

(A) $\frac{1}{18}\left[9-2\cos^6\theta-3\cos^4\theta-6\cos^2\theta\right]^{\frac{3}{2}}+c$

(B) $\frac{1}{18}\left[11-18\sin^2\theta+9\sin^4\theta-2\sin^6\theta\right]^{\frac{3}{2}}+c$

(C) $\frac{1}{18}\left[11-18\cos^2\theta+9\cos^4\theta-2\cos^6\theta\right]^{\frac{3}{2}}+c$

(D) $\frac{1}{18}\left[9-2\sin^6\theta-3\sin^4\theta-6\sin^2\theta\right]^{\frac{3}{2}}+c$

Open solution

Q33. JEE Main 2021 (Online) 24th February Morning Shift

If $\int \frac{\cos x-\sin x}{\sqrt{8-\sin 2x}}dx = a\sin^{-1}\left(\frac{\sin x+\cos x}{b}\right)+c$, where c is a constant of integration, then the ordered pair (a, b) is equal to :

(A) (-1, 3) (B) (1, 3) (C) (1, -3) (D) (3, 1)

Open solution

Q34. JEE Main 2020 (Online) 5th September Evening Slot

If

$\int \frac{\cos\theta}{5+7\sin\theta-2\cos^2\theta}d\theta = A\log_e|B(\theta)|+C$,

where C is a constant of integration, then $\frac{B(\theta)}{A}$ can be :

(A) $\frac{2\sin\theta+1}{5(\sin\theta+3)}$ (B) $\frac{2\sin\theta+1}{\sin\theta+3}$

(C) $\frac{5(2\sin\theta+1)}{\sin\theta+3}$ (D) $\frac{5(\sin\theta+3)}{2\sin\theta+1}$

Open solution

Q35. JEE Main 2020 (Online) 5th September Morning Slot

If

$\int \left(e^{2x}+2e^x-e^{-x}-1\right)e^{(e^x+e^{-x})}dx = g(x)e^{(e^x+e^{-x})}+c$

where c is a constant of integration, then g(0) is equal to :

(A) 1 (B) 2 (C) e (D) e^2

Open solution

Q36. JEE Main 2020 (Online) 4th September Morning Slot

Let $f(x)=\int \frac{\sqrt{x}}{(1+x)^2}dx\ (x\geq 0)$. Then f(3) − f(1) is equal to :

(A) $-\frac{\pi}{12}+\frac{1}{2}+\frac{\sqrt{3}}{4}$ (B) $\frac{\pi}{12}+\frac{1}{2}-\frac{\sqrt{3}}{4}$

(C) $-\frac{\pi}{6}+\frac{1}{2}+\frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6}+\frac{1}{2}-\frac{\sqrt{3}}{4}$

Open solution

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Q37. JEE Main 2020 (Online) 4th September Morning Slot

The integral $\int \left(\frac{x}{x\sin x+\cos x}\right)^2 dx$ is equal to
(where C is a constant of integration):

(A) $\sec x - \frac{x\tan x}{x\sin x+\cos x} + C$

(B) $\sec x + \frac{x\tan x}{x\sin x+\cos x} + C$

(C) $\tan x - \frac{x\sec x}{x\sin x+\cos x} + C$

(D) $\tan x + \frac{x\sec x}{x\sin x+\cos x} + C$

Open solution

Q38. JEE Main 2020 (Online) 3rd September Evening Slot

If $\int \sin^{-1}\left(\sqrt{\frac{x}{1+x}}\right)dx$ = A(x)$\tan^{-1}\left(\sqrt{x}\right)$ + B(x) + C,
where C is a constant of integration, then the ordered pair (A(x), B(x)) can be :

(A) (x + 1, -$\sqrt{x}$)

(B) (x + 1, $\sqrt{x}$)

(C) (x - 1, -$\sqrt{x}$)

(D) (x - 1, $\sqrt{x}$)

Open solution

Q39. JEE Main 2020 (Online) 9th January Evening Slot

If $\int \frac{d\theta}{\cos^2\theta(\tan 2\theta+\sec 2\theta)} = \lambda\tan\theta + 2\log_e|f(\theta)| + C$

where C is a constant of integration, then the ordered pair (λ, f(θ)) is equal to :

(A) (−1, 1 − tanθ)

(B) (1, 1 + tanθ)

(C) (−1, 1 + tanθ)

(D) (1, 1 − tanθ)

Open solution

Q40. JEE Main 2020 (Online) 9th January Morning Slot

If f'(x) = $\tan^{-1}$(secx + tanx), $-\frac{\pi}{2} < x < \frac{\pi}{2}$,
and f(0) = 0, then f(1) is equal to :

(A) $\frac{1}{4}$

(B) $\frac{\pi-1}{4}$

(C) $\frac{\pi+1}{4}$

(D) $\frac{\pi+2}{4}$

Open solution

Q41. JEE Main 2020 (Online) 9th January Morning Slot

The integral $\int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}}$ is equal to :
(where C is a constant of integration)

(A) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}} + C$

(B) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$

(C) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{\frac{13}{7}} + C$

(D) $-\left(\frac{x-3}{x+4}\right)^{-\frac{1}{7}} + C$

Open solution

Q42. JEE Main 2020 (Online) 8th January Morning Slot

If $\int \frac{\cos x dx}{\sin^3 x\left(1+\sin^6 x\right)^{2/3}} = f(x)\left(1+\sin^6 x\right)^{1/\lambda} + c$

where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to

(A) $\frac{9}{8}$

(B) 2

(C) -2

(D) $-\frac{9}{8}$

Open solution

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Q43. JEE Main 2019 (Online) 12th April Evening Slot

Let $\alpha \in \left(0, \frac{\pi}{2}\right)$ be fixed. If the integral

$\int \frac{\tan x+\tan \alpha}{\tan x-\tan \alpha} dx$ = A(x) cos 2α + B(x) sin 2α + C, where C is a

constant of integration, then the functions A(x) and B(x) are respectively :

(A) $x-\alpha$ and $\log_e |\cos(x-\alpha)|$

(B) $x+\alpha$ and $\log_e |\sin(x-\alpha)|$

(C) $x+\alpha$ and $\log_e |\sin(x+\alpha)|$

(D) $x-\alpha$ and $\log_e |\sin(x-\alpha)|$

Open solution

Q44. JEE Main 2019 (Online) 12th April Morning Slot

The integral $\int \frac{2x^3-1}{x^4+x} dx$ is equal to :
(Here C is a constant of integration)

(A) $\log_e \frac{|x^3+1|}{x^2} + C$

(B) $\frac{1}{2}\log_e \frac{|x^3+1|}{x^2} + C$

(C) $\log_e \left|\frac{x^3+1}{x}\right| + C$

(D) $\frac{1}{2}\log_e \frac{(x^3+1)^2}{|x^3|} + C$

Open solution

Q45. JEE Main 2019 (Online) 10th April Evening Slot

If $\int x^5 e^{-x^2} dx = g(x)e^{-x^2} + c$, where c is a constant of integration, then $g(-1)$ is equal to :

(A) 1 (B) - 1 (C) $-\frac{5}{2}$ (D) $-\frac{1}{2}$

Open solution

Q46. JEE Main 2019 (Online) 10th April Morning Slot

If $\int \frac{dx}{(x^2-2x+10)^2} = A\left(\tan^{-1}\left(\frac{x-1}{3}\right) + \frac{f(x)}{x^2-2x+10}\right) + C$

where C is a constant of integration then :

(A) A = $\frac{1}{54}$ and f(x) = 9(x−1)^2

(B) A = $\frac{1}{54}$ and f(x) = 3(x−1)

(C) A = $\frac{1}{81}$ and f(x) = 3(x−1)

(D) A = $\frac{1}{27}$ and f(x) = 9(x−1)^2

Open solution

Q47. JEE Main 2019 (Online) 9th April Evening Slot

$\int e^{\sec x}(\sec x \tan x f(x) + \sec x \tan x + \sec^2 x)dx$
= $e^{\sec x}$f(x) + C then a possible choice of f(x) is :-

(A) x sec x + tan x + 1/2

(B) sec x + xtan x - 1/2

(C) sec x - tan x - 1/2

(D) sec x + tan x + 1/2

Open solution

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Q48. JEE Main 2019 (Online) 9th April Morning Slot

The integral $\int \sec^{2/3} x \operatorname{cosec}^{4/3} x \, dx$ is equal to (Hence C is a constant of integration)

(A) -3/4 $\tan^{-4/3} x$ + C

(B) $3\tan^{-1/3} x$ + C

(C) $-3\cot^{-1/3} x$ + C

(D) $-3\tan^{-1/3} x$ + C

Open solution

Q49. JEE Main 2019 (Online) 8th April Evening Slot

If $\int \frac{dx}{x^3(1+x^6)^{2/3}} = xf(x)(1+x^6)^{\frac{1}{3}} + C$

where C is a constant of integration, then the function f(x) is equal to

(A) $\frac{3}{x^2}$

(B) $-\frac{1}{6x^3}$

(C) $-\frac{1}{2x^3}$

(D) $-\frac{1}{2x^2}$

Open solution

Q50. JEE Main 2019 (Online) 8th April Morning Slot

$\int \frac{\sin\frac{5x}{2}}{\sin\frac{x}{2}} dx$ is equal to

(where c is a constant of integration)

(A) 2x + sinx + 2sin2x + c

(B) x + 2sinx + sin2x + c

(C) x + 2sinx + 2sin2x + c

(D) 2x + sinx + sin2x + c

Open solution

Q51. JEE Main 2019 (Online) 12th January Evening Slot

The integral $\int \frac{3x^{13}+2x^{11}}{(2x^4+3x^2+1)^4} dx$ is equal to : (where C is a constant of integration)

(A) $\frac{x^{12}}{6(2x^4+3x^2+1)^3} + C$

(B) $\frac{x^4}{6(2x^4+3x^2+1)^3} + C$

(C) $\frac{x^{12}}{(2x^4+3x^2+1)^3} + C$

(D) $\frac{x^4}{(2x^4+3x^2+1)^3} + C$

Open solution

Q52. JEE Main 2019 (Online) 12th January Morning Slot

The integral $\int \cos(\log_e x)\, dx$ is equal to : (where C is a constant of integration)

(A) $\frac{x}{2}[\sin(\log_e x) - \cos(\log_e x)]$ + C

(B) $x[\cos(\log_e x) + \sin(\log_e x)]$ + C

(C) $\frac{x}{2}[\cos(\log_e x) + \sin(\log_e x)]$ + C

(D) $x[\cos(\log_e x) - \sin(\log_e x)]$ + C

Open solution

Q53. JEE Main 2019 (Online) 11th January Evening Slot

If $\int \frac{x+1}{\sqrt{2x-1}} dx$ = f(x) $\sqrt{2x-1}$ + C, where C is a constant of integration, then f(x) is equal to :

(A) $\frac{2}{3}$ (x − 4)

(B) $\frac{1}{3}$ (x + 4)

(C) $\frac{1}{3}$ (x + 1)

(D) $\frac{2}{3}$ (x + 2)

Open solution

Q54. JEE Main 2019 (Online) 11th January Morning Slot

If $\int \frac{\sqrt{1-x^2}}{x^4} dx = A(x)\left(\sqrt{1-x^2}\right)^m$ + C, for a suitable chosen integer m and a function A(x), where C is a constant of integration, then $(A(x))^m$ equals :

(A) $\frac{1}{27x^6}$

(B) $\frac{-1}{27x^9}$

(C) $\frac{1}{9x^4}$

(D) $\frac{1}{3x^3}$

Open solution

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Q55. JEE Main 2019 (Online) 10th January Evening Slot

If $\int x^5.e^{-4x^3}\,dx = \frac{1}{48}e^{-4x^3} f(x) + C$, where C is a constant of inegration, then f(x) is equal to -

(A) $-2x^3 - 1$ (B) $-2x^3 + 1$ (C) $4x^3 + 1$ (D) $-4x^3 - 1$

Open solution

Q56. JEE Main 2019 (Online) 10th January Morning Slot

Let n $\geq$ 2 be a natural number and $0 < \theta < \frac{\pi}{2}$. Then $\int \frac{(\sin^n\theta - \sin\theta)^{1/n}\cos\theta}{\sin^{n+1}\theta}\,d\theta$ is equal to - (where C is a constant of integration)

(A) $\frac{n}{n^2-1}\left(1+\frac{1}{\sin^{n-1}\theta}\right)^{\frac{n+1}{n}} + C$

(B) $\frac{n}{n^2-1}\left(1-\frac{1}{\sin^{n+1}\theta}\right)^{\frac{n+1}{n}} + C$

(C) $\frac{n}{n^2-1}\left(1-\frac{1}{\sin^{n-1}\theta}\right)^{\frac{n+1}{n}} + C$

(D) $\frac{n}{n^2+1}\left(1-\frac{1}{\sin^{n-1}\theta}\right)^{\frac{n+1}{n}} + C$

Open solution

Q57. JEE Main 2019 (Online) 9th January Evening Slot

If $f(x) = \int \frac{5x^8+7x^6}{(x^2+1+2x^7)^2}\,dx,\ (x \geq 0),$

$f(0) = 0,$ then the value of $f(1)$ is :

(A) $-\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Open solution

Q58. JEE Main 2019 (Online) 9th January Morning Slot

For $x^2 \neq n\pi + 1$, n $\in$ N (the set of natural numbers), the integral

$\int x\sqrt{\frac{2\sin(x^2-1)-\sin 2(x^2-1)}{2\sin(x^2-1)+\sin 2(x^2-1)}}\,dx$ is equal to :

(where c is a constant of integration)

(A) $\log_e\left|\frac{1}{2}\sec^2(x^2-1)\right| + c$

(B) $\frac{1}{2}\log_e\left|\sec(x^2-1)\right| + c$

(C) $\frac{1}{2}\log_e\left|\sec^2\left(\frac{x^2-1}{2}\right)\right| + c$

(D) $\log_e\left|\sec\left(\frac{x^2-1}{2}\right)\right| + c$

Open solution

Q59. JEE Main 2018 (Online) 16th April Morning Slot

If $\int \frac{\tan x}{1+\tan x+\tan^2 x}\,dx = x - \frac{K}{\sqrt{A}}\tan^{-1}\left(\frac{K\tan x+1}{\sqrt{A}}\right) + C, (C$ is a constant of integration) then the ordered pair (K, A) is equal to :

(A) (2, 1) (B) $(-2, 3)$ (C) (2, 3) (D) $(-2, 1)$

Open solution

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Q60. JEE Main 2018 (Offline)

Open solution

The integral

$$\int \frac{\sin^2 x \cos^2 x}{\left(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x\right)^2} dx$$

is equal to

(A) $\frac{-1}{1+\cot^3 x} + C$

(B) $\frac{1}{3(1+\tan^3 x)} + C$

(C) $\frac{-1}{3(1+\tan^3 x)} + C$

(D) $\frac{1}{1+\cot^3 x} + C$

Q61. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

If $\int \frac{2x+5}{\sqrt{7-6x-x^2}}\, dx = A\sqrt{7-6x-x^2} + B\sin^{-1}\left(\frac{x+3}{4}\right) + C$
(where C is a constant of integration), then the ordered pair (A, B) is equal to :

(A) (2, 1) (B) (− 2, −1) (C) (− 2, 1) (D) (2, −1)

Q62. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

If $f\left(\frac{x-4}{x+2}\right) = 2x + 1$, (x ∈ **R** −{1, − 2}), then $\int f(x)dx$ is equal to :
(where C is a constant of integration)

(A) 12 $\log_e$ | 1 − x | + 3x + C

(B) − 12 $\log_e$ | 1 − x | − 3x + C

(C) 12 $\log_e$ | 1 − x | − 3x + C

(D) − 12 $\log_e$ | 1 − x | + 3x + C

Q63. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

If $f\left(\frac{3x-4}{3x+4}\right)$ = x + 2, $x \neq -\frac{4}{3}$, and

$\int f(x)\, dx = A \log|1 - x| + Bx + C$,

then the ordered pair (A, B) is equal to :

(where C is a constant of integration)

(A) $\left(\frac{8}{3}, \frac{2}{3}\right)$

(B) $\left(-\frac{8}{3}, \frac{2}{3}\right)$

(C) $\left(-\frac{8}{3}, -\frac{2}{3}\right)$

(D) $\left(\frac{8}{3}, -\frac{2}{3}\right)$

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Q64. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

The integral

$$\int \sqrt{1+2\cot x(\operatorname{cosec} x+\cot x)}\ dx$$

$\left(0<x<\frac{\pi}{2}\right)$ is equal to :

(where C is a constant of integration)

(A) 4 log(sin $\frac{x}{2}$) + C

(B) 2 log(sin $\frac{x}{2}$) + C

(C) 2 log(cos $\frac{x}{2}$) + C

(D) 4 log(cos $\frac{x}{2}$) + C

Q65. JEE Main 2017 (Offline)

Open solution

Let $I_n=\int \tan^n x\,dx,\ (n>1)$.

If I_4+I_6 = $a\tan^5 x+bx^5+C$, where C is a constant of integration,

then the ordered pair (a,b) is equal to

(A) $\left(\frac{1}{5},0\right)$

(B) $\left(\frac{1}{5},-1\right)$

(C) $\left(-\frac{1}{5},0\right)$

(D) $\left(-\frac{1}{5},1\right)$

Q66. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

The integral $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$ is equal to :

(where C is a constant of integration.)

(A) $-2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}+C$

(B) $-2\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}+C$

(C) $-\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}+C$

(D) $2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}+C$

Q67. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

If $\int \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}=(\tan x)^A+C(\tan x)^B+k,$

where k is a constant of integration, then A + B +C equals :

(A) $\frac{21}{5}$

(B) $\frac{16}{5}$

(C) $\frac{7}{10}$

(D) $\frac{27}{10}$

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Q68. JEE Main 2016 (Offline)

The integral $\int \frac{2x^{12}+5x^9}{(x^5+x^3+1)^3}dx$ is equal to :

(A) $\frac{x^5}{2(x^5+x^3+1)^2}+C$

(B) $\frac{-x^{10}}{2(x^5+x^3+1)^2}+C$

(C) $\frac{-x^5}{(x^5+x^3+1)^2}+C$

(D) $\frac{x^{10}}{2(x^5+x^3+1)^2}+C$

Open solution

Q69. JEE Main 2015 (Online) 11th April Morning Slot

If $\int \frac{\log\left(t+\sqrt{1+t^2}\right)}{\sqrt{1+t^2}}dt = \frac{1}{2}(g(t))^2 + C$,

where C is a constant, then g(2) is equal to :

(A) $2\log(2+\sqrt{5})$

(B) $\log(2+\sqrt{5})$

(C) $\frac{1}{\sqrt{5}}\log(2+\sqrt{5})$

(D) $\frac{1}{2}\log(2+\sqrt{5})$

Open solution

Q70. JEE Main 2015 (Online) 10th April Morning Slot

The integral $\int \frac{dx}{(x+1)^{\frac{3}{4}}(x-2)^{\frac{5}{4}}}$ is equal to :

(A) $4\left(\frac{x+1}{x-2}\right)^{\frac{1}{4}}+C$

(B) $4\left(\frac{x-2}{x+1}\right)^{\frac{1}{4}}+C$

(C) $-\frac{4}{3}\left(\frac{x+1}{x-2}\right)^{\frac{1}{4}}+C$

(D) $-\frac{4}{3}\left(\frac{x-2}{x+1}\right)^{\frac{1}{4}}+C$

Open solution

Q71. JEE Main 2015 (Offline)

The integral $\int \frac{dx}{x^2(x^4+1)^{3/4}}$ equals :

(A) $-\left(x^4+1\right)^{\frac{1}{4}}+c$

(B) $-\left(\frac{x^4+1}{x^4}\right)^{\frac{1}{4}}+c$

(C) $\left(\frac{x^4+1}{x^4}\right)^{\frac{1}{4}}+c$

(D) $\left(x^4+1\right)^{\frac{1}{4}}+c$

Open solution

Q72. JEE Main 2014 (Online) 19th April Morning Slot

If m is a non - zero number and $\int \frac{x^{5m-1}+2x^{4m-1}}{(x^{2m}+x^m+1)^3}dx = f(x)+c$,

then f(x) is :

(A) $\frac{x^{5m}}{2m(x^{2m}+x^m+1)^2}$

(B) $\frac{x^{4m}}{2m(x^{2m}+x^m+1)^2}$

(C) $\frac{2m\left(x^{5m}+x^{4m}\right)}{(x^{2m}+x^m+1)^2}$

(D) $\frac{\left(x^{5m}-x^{4m}\right)}{2m(x^{2m}+x^m+1)^2}$

Open solution

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Q73. JEE Main 2014 (Online) 12th April Morning Slot

The integral $\int \frac{\sin^2 x \cos^2 x}{\left(\sin^3 x + \cos^3 x\right)^2} dx$ is equal to :

(A) $\frac{1}{\left(1+\cot^3 x\right)} + c$

(B) $-\frac{1}{3\left(1+\tan^3 x\right)} + c$

(C) $\frac{\sin^3 x}{\left(1+\cos^3 x\right)} + c$

(D) $-\frac{\cos^3 x}{3\left(1+\sin^3 x\right)} + c$

Open solution

Q74. JEE Main 2014 (Online) 11th April Morning Slot

The integral $\int x \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) dx \ (x > 0)$ is equal to :

(A) $-x + (1 + x^2)\tan^{-1} x + c$

(B) $x - (1 + x^2)\cot^{-1} x + c$

(C) $-x + (1 + x^2)\cot^{-1} x + c$

(D) $x - (1 + x^2)\tan^{-1} x + c$

Open solution

Q75. JEE Main 2014 (Online) 9th April Morning Slot

$\int \frac{\sin^8 x - \cos^8 x}{\left(1 - 2\sin^2 x \cos^2 x\right)} dx$ is equal to :

(A) $\frac{1}{2}$ sin 2x + c

(B) $-\frac{1}{2}$ sin 2x + c

(C) $-\frac{1}{2}$ sin x + c

(D) $-\sin^2 x + c$

Open solution

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Q76. JEE Main 2014 (Offline)

The integral $\int \left(1+x-\frac{1}{x}\right)e^{x+\frac{1}{x}}dx$ is equal to

Open solution

(A)

$$\left(x+1\right)e^{x+\frac{1}{x}}+c$$

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(B)

$$-xe^{x+\frac{1}{x}}+c$$

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(C)

$$\left(x-1\right)e^{x+\frac{1}{x}}+c$$

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(D)

$$xe^{x+\frac{1}{x}}+c$$

ExamSide.Com

Q77. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

If the integral

$$\int \frac{\cos 8x+1}{\cot 2x-\tan 2x}dx = A\cos 8x + k,$$

where k is an arbitrary constant, then A is equal to :

(A) $-\frac{1}{16}$ (B) $\frac{1}{16}$ (C) $\frac{1}{8}$ (D) $-\frac{1}{8}$

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Q78. JEE Main 2013 (Online) 23th April Morning Slot

The integral $\int \frac{xdx}{2-x^2+\sqrt{2-x^2}}$ equals :

(A) $\log\left|1+\sqrt{2+x^2}\right|+c$

(B) $-\log\left|1+\sqrt{2-x^2}\right|+c$

(C) $-x\log\left|1-\sqrt{2-x^2}\right|+c$

(D) $x\log\left|1-\sqrt{2+x^2}\right|+c$

Open solution

Q79. JEE Main 2013 (Online) 22th April Morning Slot

If $\int \frac{x^2-x+1}{x^2+1}e^{\cot^{-1}x}\,dx = A(x)e^{\cot^{-1}x}+C$, then A(x) is equal to :

(A) $-$ x

(B) x

(C) $\sqrt{1-x}$

(D) $\sqrt{1+x}$

Open solution

Q80. JEE Main 2013 (Online) 9th April Morning Slot

If $\int \frac{dx}{x+x^7} = p(x)$ then, $\int \frac{x^6}{x+x^7}\,dx$ is equal to :

(A) $\ln|x|-$ p(x) + c

(B) $\ln|x|$ + p(x) + c

(C) x $-$ p(x) + c

(D) x + p(x) + c

Open solution

Q81. JEE Main 2013 (Offline)

If $\int f(x)dx = \psi(x)$, then $\int x^5 f\left(x^3\right)dx$ is equal to

(A) $\frac{1}{3}\left[x^3\psi\left(x^3\right)-\int x^2\psi\left(x^3\right)dx\right]+C$

(B) $\frac{1}{3}x^3\psi\left(x^3\right)-3\int x^3\psi\left(x^3\right)dx+C$

(C) $\frac{1}{3}x^3\psi\left(x^3\right)-\int x^2\psi\left(x^3\right)dx+C$

(D) $\frac{1}{3}\left[x^3\psi\left(x^3\right)-\int x^3\psi\left(x^3\right)dx\right]+C$

Open solution

Q82. AIEEE 2012 (Online) 26th May

$f(x) = \int \frac{dx}{\sin^6 x}$ is a polynomial of degree

(A) 5 in $\cot x$

(B) 5 in $\tan x$

(C) 3 in $\tan x$

(D) 3 in $\cot x$

Open solution

Q83. AIEEE 2012 (Online) 19th May

If $f(x) = \int \left(\frac{x^2+\sin^2 x}{1+x^2}\right)\sec^2 x\,dx$ and $f(0) = 0$, then $f(1)$ equals

(A) $\tan 1 - \frac{\pi}{4}$

(B) $\tan 1 + 1$

(C) $\frac{\pi}{4}$

(D) $1 - \frac{\pi}{4}$

Open solution

Q84. AIEEE 2012 (Online) 12th May

The integral of $\frac{x^2-x}{x^3-x^2+x-1}$ w.r.t. x is

(A) $\frac{1}{2}\log\left(x^2+1\right)+c$

(B) $\frac{1}{2}\log\left|x^2-1\right|+c$

(C) $\log\left(x^2+1\right)+c$

(D) $\log\left|x^2-1\right|+c$

Open solution

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Q85. AIEEE 2012 (Online) 7th May

Let $f(x)$ be an indefinite integral of $\cos^3 x$.
Statement 1: $f(x)$ is a periodic function of period π.
Statement 2: $\cos^3 x$ is a periodic function.

(A) Statement 1 is true, Statement 2 is false.

(B) Both the Statements are true, but Statement 2 is not the correct explanation of Statement 1.

(C) Both the Statements are true, and Statement 2 is correct explanation of Statement 1.

(D) Statement 1 is false, Statement 2 is true.

Open solution

Q86. AIEEE 2012

If the $\int \frac{5\tan x}{\tan x-2}dx = x + a\ \ln|\sin x - 2\cos x| + k$, then a is equal to :

(A) -1 (B) -2 (C) 1 (D) 2

Open solution

Q87. AIEEE 2008

The value of $\sqrt{2}\int \frac{\sin x dx}{\sin\left(x-\frac{\pi}{4}\right)}$ is

(A) $x + \log\left|\cos\left(x-\frac{\pi}{4}\right)\right| + c$ (B) $x - \log\left|\sin\left(x-\frac{\pi}{4}\right)\right| + c$

(C) $x + \log\left|\sin\left(x-\frac{\pi}{4}\right)\right| + c$ (D) $x - \log\left|\cos\left(x-\frac{\pi}{4}\right)\right| + c$

Open solution

Q88. AIEEE 2007

$\int \frac{dx}{\cos x+\sqrt{3}\sin x}$ equals

(A) $\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$

(B) $\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

(C) $\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$

(D) $\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

Open solution

Q89. AIEEE 2005

$\int\left\{\frac{(\log x-1)}{1+(\log x)^2}\right\}^2 dx$ is equal to

(A) $\frac{\log x}{(\log x)^2+1}+C$ (B) $\frac{x}{x^2+1}+C$

(C) $\frac{xe^x}{1+x^2}+C$ (D) $\frac{x}{(\log x)^2+1}+C$

Open solution

Q90. AIEEE 2004

If $\int \frac{\sin x}{\sin(x-\alpha)}dx = Ax + B\log\sin(x-\alpha), +C$, then value of (A, B) is

(A) $(-\cos\alpha, \sin\alpha)$ (B) $(\cos\alpha, \sin\alpha)$

(C) $(-\sin\alpha, \cos\alpha)$ (D) $(\sin\alpha, \cos\alpha)$

Open solution

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Q91. AIEEE 2004

$\int \frac{dx}{\cos x - \sin x}$ is equal to

(A) $\frac{1}{\sqrt{2}} \log \left|\tan\left(\frac{x}{2} + \frac{3\pi}{8}\right)\right| + C$

(B) $\frac{1}{\sqrt{2}} \log \left|\cot\left(\frac{x}{2}\right)\right| + C$

(C) $\frac{1}{\sqrt{2}} \log \left|\tan\left(\frac{x}{2} - \frac{3\pi}{8}\right)\right| + C$

(D) $\frac{1}{\sqrt{2}} \log \left|\tan\left(\frac{x}{2} - \frac{\pi}{8}\right)\right| + C$

Open solution

Numerical 14 questions

Q92. JEE Main 2026 (Online) 22nd January Morning Shift

If $\int (\sin x)^{\frac{-11}{2}} (\cos x)^{\frac{-5}{2}} dx = -\frac{p_1}{q_1}(\cot x)^{\frac{9}{2}} - \frac{p_2}{q_2}(\cot x)^{\frac{5}{2}} - \frac{p_3}{q_3}(\cot x)^{\frac{1}{2}} + \frac{p_4}{q_4}(\cot x)^{\frac{-3}{2}} + C$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{15 p_1 p_2 p_3 p_4}{q_1 q_2 q_3 q_4}$ is equal to ____

Open solution

Q93. JEE Main 2025 (Online) 7th April Evening Shift

If $\int \left(\frac{1}{x} + \frac{1}{x^3}\right) \left(\sqrt[23]{3x^{-24} + x^{-26}}\right) dx = -\frac{\alpha}{3(\alpha+1)} \left(3x^{\beta} + x^{\gamma}\right)^{\frac{\alpha+1}{\alpha}} + C, x > 0, (\alpha, \beta, \gamma \in \mathbf{Z})$, where C is the constant of integration, then $\alpha + \beta + \gamma$ is equal to _____.

Open solution

Q94. JEE Main 2025 (Online) 4th April Evening Shift

If $\int \frac{\left(\sqrt{1+x^2}+x\right)^{10}}{\left(\sqrt{1+x^2}-x\right)^{9}} \, dx = \frac{1}{m}\left(\left(\sqrt{1+x^2}+x\right)^{n}\left(n\sqrt{1+x^2}-x\right)\right) + C$ where C is the constant of integration and $m, n \in \mathbf{N}$, then $m + n$ is equal to _____ .

Open solution

Q95. JEE Main 2025 (Online) 24th January Evening Shift

If $\int \frac{2x^2+5x+9}{\sqrt{x^2+x+1}} \, dx = x\sqrt{x^2+x+1} + \alpha\sqrt{x^2+x+1} + \beta \log_e \left|x + \frac{1}{2} + \sqrt{x^2+x+1}\right| + C$, where C is the constant of integration, then $\alpha + 2\beta$ is equal to _____ .

Open solution

Q96. JEE Main 2024 (Online) 8th April Evening Shift

If $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}} \, dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^{B} + C$, where C is the constant of integration, then the value of $\alpha + \beta + 20AB$ is _____.

Open solution

Q97. JEE Main 2024 (Online) 4th April Evening Shift

If $\int \operatorname{cosec}^5 x dx = \alpha \cot x \operatorname{cosec} x \left(\operatorname{cosec}^2 x + \frac{3}{2}\right) + \beta \log_e \left|\tan \frac{x}{2}\right| + C$ where $\alpha, \beta \in \mathbb{R}$ and C is the constant of integration, then the value of $8(\alpha + \beta)$ equals _____.

Open solution

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Q98. JEE Main 2023 (Online) 15th April Morning Shift

Let $f(x)=\int \frac{dx}{(3+4x^2)\sqrt{4-3x^2}}, |x|<\frac{2}{\sqrt{3}}$. If $f(0)=0$ and $f(1)=\frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right), \alpha,\beta>0$, then $\alpha^2+\beta^2$ is equal to _____.

Open solution

Q99. JEE Main 2023 (Online) 12th April Morning Shift

Let $I(x)=\int \sqrt{\frac{x+7}{x}}\,dx$ and $I(9)=12+7\log_e 7$. If $I(1)=\alpha+7\log_e(1+2\sqrt{2})$, then α^4 is equal to _____.

Open solution

Q100. JEE Main 2023 (Online) 30th January Evening Shift

If $\int \sqrt{\sec 2x-1}dx=\alpha\log_e\left|\cos 2x+\beta+\sqrt{\cos 2x\left(1+\cos\frac{1}{\beta}x\right)}\right|+$ constant, then $\beta-\alpha$ is equal to _____.

Open solution

Q101. JEE Main 2021 (Online) 31st August Evening Shift

If $\int \frac{\sin x}{\sin^3 x+\cos^3 x}dx=$

$\alpha\log_e|1+\tan x|+\beta\log_e|1-\tan x+\tan^2 x|+\gamma\tan^{-1}\left(\frac{2\tan x-1}{\sqrt{3}}\right)+C$, where C is constant of integration, then the value of $18(\alpha+\beta+\gamma^2)$ is _____.

Open solution

Q102. JEE Main 2021 (Online) 27th August Evening Shift

If $\int \frac{2e^x+3e^{-x}}{4e^x+7e^{-x}}dx=\frac{1}{14}(ux+v\log_e(4e^x+7e^{-x}))+C$, where C is a constant of integration, then u + v is equal to _____.

Open solution

Q103. JEE Main 2021 (Online) 27th August Morning Shift

If $\int \frac{dx}{(x^2+x+1)^2}=a\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+b\left(\frac{2x+1}{x^2+x+1}\right)+C$, x > 0 where C is the constant of integration, then the value of $9\left(\sqrt{3}a+b\right)$ is equal to _____.

Open solution

Q104. JEE Main 2021 (Online) 18th March Morning Shift

If $f(x)=\int \frac{5x^8+7x^6}{(x^2+1+2x^7)^2}dx, (x\geq 0), f(0)=0$ and $f(1)=\frac{1}{K}$, then the value of K is

Open solution

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Q105. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

For real numbers α, β, γ and δ, if

$$\int \frac{(x^2-1)+\tan^{-1}\left(\frac{x^2+1}{x}\right)}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)}dx$$

$$= \alpha\log_e\left(\tan^{-1}\left(\frac{x^2+1}{x}\right)\right) + \beta\tan^{-1}\left(\frac{\gamma(x^2-1)}{x}\right) + \delta\tan^{-1}\left(\frac{x^2+1}{x}\right) + C$$

where C is an arbitrary constant, then the value of $10(\alpha + \beta\gamma + \delta)$ is equal to _____.

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **B**	4. **C**	5. **D**
6. **C**	7. **A**	8. **A**	9. **B**	10. **A**
11. **A**	12. **D**	13. **D**	14. **C**	15. **B**
16. **D**	17. **A**	18. **D**	19. **D**	20. **B**
21. **C**	22. **A**	23. **D**	24. **A**	25. **A**
26. **B**	27. **A**	28. **D**	29. **C**	30. **A**
31. **B**	32. **C**	33. **B**	34. **C**	35. **B**
36. **B**	37. **C**	38. **A**	39. **C**	40. **C**
41. **B**	42. **C**	43. **D**	44. **C**	45. **C**
46. **B**	47. **D**	48. **D**	49. **C**	50. **B**
51. **A**	52. **C**	53. **B**	54. **B**	55. **D**
56. **C**	57. **D**	58. **D**	59. **C**	60. **C**
61. **B**	62. **B**	63. **B**	64. **B**	65. **A**
66. **B**	67. **B**	68. **D**	69. **B**	70. **C**
71. **B**	72. **B**	73. **B**	74. **A**	75. **B**
76. **D**	77. **A**	78. **B**	79. **B**	80. **A**
81. **C**	82. **A**	83. **A**	84. **A**	85. **D**
86. **D**	87. **C**	88. **C**	89. **D**	90. **B**
91. **A**				

Numerical

92. **16**	93. **19**	94. **379**	95. **16**	96. **7**
97. **1**	98. **28**	99. **64**	100. **1**	101. **3**
102. **7**	103. **15**	104. **4**	105. **6**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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